

Population pharmacokinetic and exposure-response analysis of eptinezumab in the treatment of episodic and chronic migraine.

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Eptinezumab is a humanized mAb that targets calcitonin gene-related peptide and is under regulatory review for the prevention of episodic and chronic migraine (EM, CM). It is important to determine whether exposures achieved with intravenous (IV) administration of eptinezumab achieve desired pharmacologic effects. Population pharmacokinetics, including dose- and exposure-response analyses, were performed using patient-level data from the eptinezumab clinical trial program with IV doses ranging from 10 to 1000 mg in pharmacokinetic analyses or 10 to 300 mg in phase 2/3 clinical studies in patients with EM or CM. Exposure-response analysis explored the relationship between eptinezumab exposure metrics and efficacy parameters including monthly migraine days. The pharmacokinetic profile of eptinezumab was characterized by rapid attainment of maximum plasma concentration (ie, end of IV administration) and a terminal half-life of 27 days. Covariate analysis found that patient characteristics had no clinically significant effects on pharmacokinetic parameters and were insufficient to influence dosing. Dose- and exposure-response analyses found exposure with single doses ≥ 100 mg was associated with greater efficacy compared with doses ≤ 30 mg and a plateau of effect between 100 and 300 mg. A saturable inhibitory Emax model found the exposure over 12 weeks produced by single-dose eptinezumab 100 and 300 mg exceeded the exposure estimates required to achieve 90% of the maximal efficacy (EC90). This pharmacokinetic analysis of eptinezumab supports dosing every 12 weeks with no adjustment for patient characteristics, including exposures associated with 100- or 300-mg doses producing optimal efficacy effects. The similar efficacy profiles support 100 mg as the lowest effective dose of eptinezumab.

Eptinezumab for the preventive treatment of migraine.

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Our knowledge of the pathophysiology of migraine and the molecules implicated in the disorder have evolved over time. Among these, calcitonin gene-related peptide has shown a crucial role that led to the development of therapies specifically targeting the molecule. Four monoclonal antibodies targeting the calcitonin gene-related peptide pathway are currently available after the US FDA approval of eptinezumab for the indication of migraine prevention. This is the only one of the class to be administered intravenously. The pharmacology of eptinezumab and the four studies that led to the approval, two Phase II and two Phase III clinical trials, are reviewed in this paper. Eptinezumab has demonstrated efficacy, tolerability and safety in patients with episodic and chronic migraine. Studies including migraineurs who have failed previous preventives, and comparison with other options administered quarterly, as well as real-world experience data will all be welcome.

The CGRP Pathway in Migraine as a Viable Target for Therapies.

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The neuropeptide calcitonin gene-related peptide is well established as a key player in the pathogenesis of migraine. Clinical studies show calcitonin gene-related peptide levels correlate with migraine attacks, and decreases in this neuropeptide can indicate antimigraine therapy effectiveness. Research has

revealed a wide distribution of expression sites for calcitonin gene-related peptide in the central and peripheral nervous system. Of these, the calcitonin gene-related peptide receptor, which binds calcitonin gene-related peptide with high affinity, has attracted growing interest as a viable target for antimigraine therapies. An incentive to pursue such research is the continuing unmet medical need of patients. Triptans have offered some clinical benefit, but many patients do not respond and these drugs have important safety considerations. Initial calcitonin gene-related peptide-focused research led to development of the "gepant" small-molecule calcitonin gene-related peptide receptor blockers. Positive efficacy reports concerning the gepants have been tempered by safety findings which led to the discontinuation of some of these agents. Currently, there is considerable excitement regarding monoclonal antibodies against calcitonin gene-related peptide (eptinezumab, galcanezumab, fremanezumab) and the calcitonin gene-related peptide receptor (erenumab). To date, these monoclonal antibodies have shown promising efficacy in clinical trials, with no major safety concerns. If ongoing long-term studies show that their efficacy can be maintained, this may herald a new era for effective antimigraine therapies.

Calcitonin gene-related peptide (CGRP)-targeted therapies as preventive and acute treatments for migraine-The monoclonal antibodies and gepants.

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Calcitonin Gene-Related Peptide (CGRP) plays a pivotal role in migraine pathophysiology. Two types of CGRP function-blocking modalities, monoclonal antibodies, and small molecules (gepants), have been developed to target the CGRP ligands and CGRP receptors. Four CGRP monoclonal antibodies have received FDA approval for the prevention of migraine: erenumab, fremanezumab, galcanezumab, and eptinezumab. Two gepants have been approved by the FDA for the acute treatment of migraine: ubrogepant and rimegepant. Multiple clinical trials of the CGRP monoclonal antibodies and gepants, and now some open-label long-term extension data, established their efficacy, safety, and tolerability. In this chapter, we summarize the major clinical trials, pharmacokinetic insights, safety and tolerability profiles, and real-world data (if available) of the CGRP monoclonal antibodies and gepants.

[Anti-calcitonin generelated peptide (CGRP) therapies for migraine].

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ANTI-CALCITONIN GENE-RELATED PEPTIDE (CGRP) THERAPIES FOR MIGRAINE. Currently, four monoclonal antibodies targeting the CGRP (calcitonin gene-related peptide) pathway have been shown to be effective as migraine prophylactics: eptinezumab, erenumab, fremanezumab and galcanezumab. Unlike the usual preventive treatments, they are administered parenterally: subcutaneously (monthly or quarterly) or by quarterly IV infusion for eptinezumab. They reduce the frequency of attacks by at least 50% in 50 to 60% of migraine patients, even in cases of failure of several conventional preventive treatments, in cases of chronic migraine and medication overuse. Their tolerance is better than that of conventional oral treatments and the discontinuation rates are very low. They can be proposed after failure of at least two conventional prophylactic treatments, in patients with at least 8 migraine days per month and without cardiovascular pathology. Indeed, these drugs present a risk in case of cardiovascular disease, by inhibiting vasodilation, and are therefore contraindicated in this population. The main limitation to the

use of these treatments in France at present is the lack of reimbursement, the cheapest molecule being available at a price of 245 € per injection.

The impact of CGRPergic monoclonal antibodies on prophylactic antimigraine therapy and potential adverse events.

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Migraine is a prevalent medical condition and the second most disabling neurological disorder. Regarding its pathophysiology, calcitonin gene-related peptide (CGRP) plays a key role, and, consequently, specific antimigraine pharmacotherapy has been designed to target this system. Hence, apart from the gepants, the recently developed monoclonal antibodies (mAbs) are a novel approach to treat this disorder. In this review we consider the current knowledge on the mechanisms of action, specificity, safety, and efficacy of the above mAbs as prophylactic antimigraine agents, and examine the possible adverse events that these agents may trigger. Antimigraine mAbs act as direct scavengers of CGRP (galcanezumab, fremanezumab, and eptinezumab) or against the CGRP receptor (erenumab). Due to their long half-lives, these molecules have revolutionized the prophylactic treatment of this neurovascular disorder. Moreover, because of their physicochemical properties, these agents are hepato-friendly and do not cross the blood-brain barrier (highlighting the relevance of peripheral mechanisms in migraine). Nevertheless, apart from potential cardiovascular side effects, the interaction with AMY1 receptors and immunogenicity induced by autoantibodies against mAbs could be a concern for the safety of long-term treatment with these molecules.

Calcitonin Gene-Related Peptide-Targeted Therapies for Migraine and Cluster Headache: A Review.

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Calcitonin gene-related peptide (CGRP) is a signaling neuropeptide released from activated trigeminal sensory afferents in headache and facial pain disorders. There are a handful of CGRP-targeted therapies currently in phase 3 studies for migraine acute treatment or prevention. Currently, 4 monoclonal antibodies targeting either the CGRP ligand or receptor are being studied for migraine prevention: ALD403 (eptinezumab), AMG 334 (erenumab), LY2951742 (galcanezumab), and TEV-48125 (fremanezumab). Meanwhile, 1 small-molecule CGRP receptor antagonist (ubrogepant, MK-1602) is currently in phase 3 studies for the acute treatment of migraine. Two of these anti-CGRP monoclonal antibodies are in clinical trials for cluster headache prevention as well. Several other small-molecular CGRP receptor antagonists are in earlier stages of development for acute migraine treatment or prevention. In this review, we will discuss the growing body of clinical trials studying CGRP-targeted therapies for migraine and cluster headache.

Progress in the Treatment of Migraine Attacks: From Traditional Approaches to Eptinezumab.

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Migraine is the second cause of disability and of lost years of healthy life worldwide. Migraine is characterized by recurrent headache attacks and accompanying disabling symptoms lasting 4-48 h. In episodic migraine, attacks occur in less than 15 days per month and in chronic migraine, in more than 15 monthly days. Whilst successful translation of pharmacological discoveries into efficacious therapeutics has been achieved in the preventative therapy of chronic migraine, treatment of acute migraine suffers the lack of effective advancements. An effective treatment affords complete freedom from pain two hours after therapy and provides the absence of the most bothersome symptom (MBS) associated with migraine after 2 h. However, available anti-migraine abortive treatments for acute attacks do not represent an effective and safe treatment for all the populations treated. In particular, the most used specific treatment is represented by triptans that offer 2-h sustained freedom from pain achieved in 18-50% of patients but they are contraindicated in coronary artery disease, stroke and peripheral vascular disease due to the vasoconstriction at the basis of their pharmacologic action. The most novel therapies, i.e., gepants and ditans, are without sufficient post-marketing data for secure use. Here, an attempt is proposed to analyse the rational basis and evidence in favour of investigating the efficacy and safety in acute migraine attacks of eptinezumab, i.e., monoclonal antibody (mAb) directed towards calcitonin gene-related peptide (CGRP) unique for intravenous infusion administration.

Monoclonal antibodies for the prevention of migraine.

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Introduction: Calcitonin Gene-Related Peptide (CGRP) plays a crucial role in migraine pathophysiology. A novel specific treatment strategy for the prevention of migraine incorporates monoclonal antibodies (mAbs) against CGRP and its canonical receptor. Eptinezumab, fremanezumab and galcanezumab block CGRP mediated effects by binding to the peptide, while erenumab blocks the CGRP receptor. **Areas covered:** Following a brief overview of pharmacological characteristics, we will review phase III trials for the use of CGRP mAbs in the prevention of episodic and chronic migraine. **Expert opinion:** All four CGRP mAbs demonstrated an excellent safety, tolerability and efficacy profile in migraine patients. Across all trials mAbs showed superior efficacy for the reduction of monthly migraine days compared to placebo with a net benefit of 2.8 days. Neither cardiovascular nor immunological safety concerns have emerged from clinical trials. Fremanezumab, galcanezumab, and erenumab are approved in the USA and Europe. Based on trial data there is no reason why these mAbs should not become first-line therapies in future. For now, we advocate for the use of mAbs in migraine prevention for patients who failed a minimum of two standard oral treatments based on the novelty and costs of this approach. mAbs are also effective in patients with medication overuse and with comorbid depression or anxiety disorders. Taken together, mAbs are likely to usher in a new era in migraine prevention and provide significant value to patients.

Pharmacologic Characterization of ALD403, a Potent Neutralizing Humanized Monoclonal Antibody Against the Calcitonin Gene-Related Peptide.

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ALD403 is a genetically engineered, humanized immunoglobulin G1 monoclonal antibody that inhibits the action of human calcitonin gene-related peptide (CGRP). Clinical trial data indicate that ALD403 is

effective as a preventive therapy for migraine and has an acceptable safety profile. For preclinical characterization of ALD403, rabbit antibodies targeting α -CGRP were humanized and modified to eliminate fragment crystallizable (Fc) γ receptor (Fc γ R) and complement interactions. The ability of ALD403 to inhibit CGRP-induced cAMP production was assessed using a cAMP bioassay (Meso Scale Discovery). The IC₅₀ for inhibition of cAMP release was 434 and 288 pM with the rabbit-human chimera antibody and the humanized ALD403, respectively. ALD403 inhibited α -CGRP binding with an IC₅₀ of 4.7×10^{-11} and 1.2×10^{-10} M for the α -CGRP and AMY1 receptors, respectively. ALD403 did not induce antibody-dependent cellular cytotoxicity or complement-dependent cytotoxicity and did not stably interact with any of the Fc γ R mediating these functions, exhibiting only weak binding to Fc γ RI. ALD403 significantly lowered capsaicin-induced blood flow responses in rodents at all time points starting at 5 minutes postapplication in a dose-dependent manner. In conclusion, ALD403 is a potent functional ligand inhibitor of α -CGRP-driven pharmacology. SIGNIFICANCE STATEMENT: α -Calcitonin gene-related peptide blockade by ALD403 was assessed via radiolabeled ligand displacement, in vitro inhibition of cell signaling, and in vivo inhibition of capsaicin-induced vasodilation. Lack of engagement of fragment crystallizable-mediated immune-effector functions by ALD403 was shown.
